Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

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We present results on the near-side of high-p_T triggered di-hadron correlations of neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K^0_S) at intermediate p_T = 2-6 GeV/c in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The near-side di-hadron correlation in ($\Delta\phi$, $\Delta\eta$) contains two structures, a peak which is narrow in $\Delta\phi$ and $\Delta\eta$ consistent with correlations due to jet fragmentation and correlation in $\Delta\phi$ which is broad in $\Delta\eta$ which is attributed dominantly to flow. The dependence of the conditional yield of the jet-like correlation on the trigger particle momentum, associated particle momentum, and centrality are presented. In addition the particle composition of the jet-like correlation is determined using identified associated particles. Data on identified triggers are qualitatively described by PYTHIA, however the composition of the jet-like correlation is not quantitatively described by PYTHIA.

PACS numbers: 25.75.-qRelativistic heavy-ion collisions 25.75.GzParticle correlations and fluctuations

I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Measurements of nuclear modification factors [1–4] have shown that the nuclear medium created is opaque to particles with large transverse momentum (p_T). Moreover, anisotropic elliptic flow measurements demonstrate that the medium exhibits partonic degrees of freedom and has properties close to those expected of a perfect fluid [5, 6].

Studies of jets in heavy ion collisions are possible through single particle measurements [1–4], di-hadron correlations [7–17], and measurements of reconstructed jets [3, 18–21]. Measurements of reconstructed jets are useful for studying higher momenta and have provided direct evidence for partonic energy loss in the medium. However, di-hadron correlations provide complementary measurements since they enable studies of intermediate momenta, studies where the jet may be so severely modified that it is no longer recognizable as a jet, and studies where the interplay between jets and the medium is important.

Properties of jets have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum at the Relativistic Heavy Ion Collider (RHIC) [7–14] and the Large Hadron Collider (LHC) [15–17]. Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed significant modification [5, 10, 11]. The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central Au+Au collisions [5, 7, 8]. In p+p and d+Au collisions, there is a peak narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to as the jet-like correlation. In Cu+Cu and Au+Au collisions this peak is observed to be broader than that in d+Au collisions, although the yields are comparable [14]. This indicates that the jet-like correlation is dominantly produced through fragmentation.

Besides the shape modifications of jet-like correlations at intermediate transverse momentum range, the production mechanism of hadrons seems to differ from simple fragmentation. Baryon production is less suppressed in central Au+Au collisions than that of mesons resulting in baryon to meson ratios enhanced in central Au+Au collisions relative to p+p and d+Au collisions [22, 23]. The measured baryon to meson ratios increase with p_T until reaching their maximum value of ≈ 3 relative to p+p collisions at $p_T \approx 3$ GeV/c in both the strange and non-strange quark sectors. A fall-off of the baryon to meson ratio is observed for $p_T > 3$ GeV/c and both the strange and non-strange baryon to meson ratios in Au+Au collisions approach the values measured in p+p collisions at $p_T \approx 6$ GeV/c.

In this paper, studies of two-particle correlations on the near-side in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment are presented. Results from two-particle correlations in pseudo-rapidity and azimuth for neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K^0_S) at intermediate p_T ($2<p_T<6$ GeV/c) in the different collision systems are compared to unidentified charged hadron correlations. Both identified trigger particles associated with charged particles (K^0_S-h, Λ-h) and unidenfitied charged trigger particles associated with identified strange particles (h-K^0_S, h-Λ) are studied. The near-side jet-like yield is studied as a function of centrality of the collision and transverse momentum of trigger and associated particles to look for possible flavor and baryon/meson. The composition of the jet-like correlation is studied using identified associated particles to investigate possible production mechanisms. Results are compared to expectations from PYTHIA.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

The results presented in this paper are based on analysis of data from d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV taken by the STAR experiment in 2003, 2005, and 2004, respectively. For d+Au collisions, the events analyzed were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located 18 m from the nominal interaction point in the Au beam direction and accepting 95±3% of the hadronic cross section [24]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) placed at forward pseudo-rapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the two ZDCs. The MB Au+Au events were obtained using a ZDC coincidence, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions. An additional online trigger for central Au+Au collisions was used. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB. The central trigger sampled the most central 12% of the total hadronic cross section.

In order to achieve a more uniform detector acceptance in Cu+Cu and Au+Au data sets, only those events with the primary vertex position along the longitudinal beam direction (z) within 30 cm of the center of the STAR detector were used for the analysis. For the d+Au collisions this vertex position selection was extended to $|z| < 50$ cm. The number of events after the vertex cuts in individual data samples is summarized in table II.

STAR [25] is a multi-purpose spectrometer with a full azimuthal coverage consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. This analysis is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [26] with a pseudo-rapidity acceptance $|\eta| < 1.5$. The TPC has in total 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements for each charged particle track. The TPC momentum resolution is determined to be $\Delta p/p \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [26]. Charged particle tracks used in this analysis were required to have at least 15 fit points in the TPC, a DCA to the primary vertex of less than 1 cm and a pseudorapidity $|\eta| < 1.0$.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	No. of events [10^6]
d+Au	0-95%	3
Cu+Cu	0-60%	38
Au+Au	0-80%	28
Au+Au	0-12%	17

We identify weakly decaying neutral strange (V^0) particles Λ, $\bar{\Lambda}$ and K^0_S by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC as described in [27]:

$$\begin{aligned} \Lambda &\rightarrow p + \pi^-, BR = (63.9 \pm 0.5)\% \\ \bar{\Lambda} &\rightarrow \bar{p} + \pi^+, BR = (63.9 \pm 0.5)\% \\ K^0_S &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\% \end{aligned} \quad (1)$$

where BR denotes the branching ratio. The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates. Reconstructed Λ and K^0_S particles are required to be within $|\eta| < 1.0$. Topological cuts are optimized for each data set and yield signal to background ratio of at $\geq$ 15:1. For the analyses presented here, no significant difference was observed between results with Λ and $\bar{\Lambda}$ trigger and associated particles so the correlations were added to increase statistics. In the remainder of the discussion the combined particles are refered to simply as Λ baryons.

III. METHOD

A. Correlation technique

The analysis in this paper follows the method in [14]. A high-p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) is formed. This distribution, $d^2N_{\rm raw}/d\Delta\phi\, d\Delta\eta$, is normalized by the number of trigger particles, $N_{\rm trigger}$, and corrected for the efficiency and acceptance of associated tracks ε:

$$\frac{d^2N}{d\Delta\phi\, d\Delta\eta}(\Delta\phi, \Delta\eta) = \frac{1}{N_{\rm trigger}} \frac{d^2N_{\rm raw}}{d\Delta\phi d\Delta\eta} \frac{1}{\varepsilon_{\rm assoc}(\phi, \eta)} \frac{1}{\varepsilon_{\rm pair}(\Delta\phi, \Delta\eta)}. \quad (2)$$

The efficiency correction $\varepsilon_{\rm assoc}(\phi, \eta)$ is a correction for the single particle reconstruction efficiency in TPC and $\varepsilon_{\rm pair}(\Delta\phi, \Delta\eta)$ is a correction for the finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$. Since the correlations are normalized per trigger, the efficiency correction is only applied for the associated particle. The fully corrected correlation functions are averaged between positive and negative $\Delta\phi$ and $\Delta\eta$ regions and are reflected about $\Delta\phi$ =0 and $\Delta\eta$ =0 in the plots.

B. Single charged track efficiency correction

For unidentified associated particles, the efficiency correction $\varepsilon_{\rm assoc}(\phi, \eta)$ is the correction for charged hadrons, identical to that applied in [14]. This single charged track reconstruction efficiency is determined as a function of p_T, η, and centrality by simulating the TPC response to a particle and embedding the simulated signals into a real event. The single charged track efficiency is found to be approximately constant for $p_T > 2$ GeV/c and ranges from around 75% for central Au+Au events to around 85% for peripheral Cu+Cu events. The efficiency for reconstructing a track in d+Au events is 89%.

For identified strange associated particles, the reconstruction efficiency $\varepsilon_{\rm assoc}(\phi,\eta)$ is determined in a similar way, but forcing the simulated particle to decay through the channel in equation 2 and then correcting for the respective branching ratio. The efficiency for Λ, $\bar{\Lambda}$, and K^0_S ranges from 8% to 15%, increasing with momentum and decreasing with system size.

The systematic uncertainty on the efficiency correction for unidentified associated hadrons, 5%, is strongly correlated across centralities and p_T bins for each data set but not between data sets. For identified associated particle ratios the systematic errors on the efficiency correction partially cancel out and are negligible compared to the statistical uncertainties.

For the inclusive spectra the feeddown correction due to secondary Λ from Ξ baryon decays is 15%, independent of p_T [28]. For identified Λ trigger particles, we assume that feeddown lambdas do not change the correlation since little trigger particle type dependence is observed in the data. For identified associated particles, we assume the same correlation between primary and secondary Λ baryons and correct the yield of Λ associated particles by reducing the yield by 15%.

C. Pair acceptance correction

The requirement of each track being measured within $|\eta| < 1.0$ in TPC, results in a limited acceptance for track pairs. In pseudorapidity the track pair geometric acceptance is $\approx$ 100% for $\Delta\eta \approx 0$, however, near $\Delta\eta$ $\approx$ 2 the acceptance is close to 0%. In azimuth the acceptance is limited by the 12 TPC sector boundaries, leading to dips in the acceptance of track pairs in $\Delta\phi$. To correct for the limited geometric acceptance, a mixed event analysis was performed using trigger particles from one event combined with associated particles from another event, as done in [7]. The position of the event vertices were required to be within 2 cm of each other along the beam axis and have the same charged particle multiplicity within 10 particles. To increase statistics of the mixed event sample, each event with a trigger particle was mixed with ten other events.

D. Track merging correction

The track merging effect in unidentified hadron (h) correlations discussed in [14] is also present for V^0-h and h-V^0 correlations. This effect leads to a dip at small $\Delta\phi$ and $\Delta\eta$ due to overlap between the trigger and associated particles. When one of the particles is a V^0, this overlap is between one of the V^0 daughter particles and

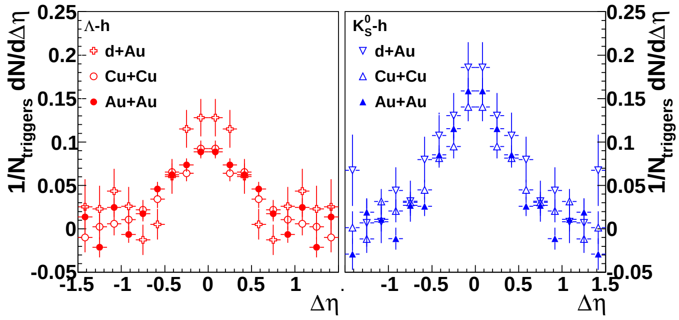

FIG. 1: Corrected correlation functions $\frac{dN_J}{d\Delta\eta}$ in $|\Delta\phi| = 1.0$ for $3 < p_T^{\rm trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\rm associated} < p_T^{\rm trigger}$ for Λ-h (left) and K^0_S-h (right) for minimum bias d+Au, 0-20% Cu+Cu, and 40-80% Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data have been reflected about $\Delta\eta = 0$.

the unidentified hadron. The size of the dip due to track merging depends strongly on the relative momenta of the particle pair. This effect is larger when the overlapping particles' momenta are closer. For V^0-h correlations, the typical associated particle momentum is approximately 1.5 GeV/c. Since the K^0_S decay is symmetric, this means that the track merging effect is greatest for K^0_S-h correlations with a trigger K^0_S momentum of approximately 3 GeV/c. In a Λ decay, the proton daughter carries more of the Λ momentum than the pion daughter. Therefore this effect is larger for Λ trigger particles with lower momenta. Since track merging affects both signal and background particles and the signal sits on top of a large combinatorial background, the effect is larger for collisions with a higher charged track multiplicity. Since the dip in V^0-h and h-V^0 correlations is the result of a V^0 daughter merged with an unidentified hadron, the dip is wider in $\Delta\phi$ and $\Delta\eta$ than in unidentified hadron correlations. When the track merging dip is present, it is corrected by fitting a Gaussian to the peak, excluding the region impacted by track merging, and using the fit to extract the yield.

E. Yield extraction

Extraction of the yield in this paper follows the method and the notation in [7, 14]. The jet-like correlation is narrow in both $\Delta\phi$ and $\Delta\eta$ and is contained within the kinematic cuts in $p_T^{\rm trigger}$ and $p_T^{\rm associated}$ used in this analysis. The di-hadron correlation from equation 2 is projected onto the $\Delta\eta$ axis:

$$\left.\frac{dN}{d\Delta\eta}\right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}. \quad (3)$$

All other correlations, including those from v_2, $V_{3\Delta}$, and higher order flow harmonics, are assumed to be independent of $\Delta\eta$ within the η acceptance of the analysis, consistent with [7, 29–31]. This means that the jet-like yield can be determined by:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left.\frac{dN}{d\Delta\eta}\right|_{-0.78,0.78} - b_{\Delta\eta} \qquad (4)$$

where $b_{\Delta\eta}$ is a constant offset determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. Sample correlations are given in Figure 1. When the dip due to track merging is negligible, the yield determined from fit is discarded to avoid any assumptions about the shape of the peak and instead we integrate equation 4 over $\Delta\eta$ using bin counting to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \, \frac{dN_J}{d\Delta\eta}(\Delta\eta) \ . \qquad (5)$$

When it is not possible, the yield from the Gaussian fit is used, excluding the region affected by track merging, and the statistical errors from the fit exceed any error from the assumption of a Gaussian shape.

IV. RESULTS

A. Correlations with identified strange trigger particles

Differences between the jet-like yield in different systems can either be due to differences in the jet spectrum contributing to the jet-like correlation or in modifications of the fragmentation functions. At the LHC fragmentation functions in Pb+Pb collisions enhanced at lower momenta relative to fragmentation functions in p+p collisions [35, 36]. If this is also true at RHIC energies and at the lower jet energies sampled by with $3 < p_T^{\mathrm{trigger}} <$ 6 GeV/c, the jet-like yield in Cu+Cu and Au+Au collisions would be expected to be enhanced rather than suppressed. However, these modifications could be dependent on the momentum or the flavor of jets sampled in the di-hadron correlation. Since no modification is observed for unidentified di-hadron correlations, this indicates that there is little modification of the fragmentation function or jet spectrum sampled in the kinematic range studied in this paper ($3 < p_T^{\mathrm{trigger}} < 6$ GeV/c, 1.5 GeV/c $< p_T^{\mathrm{associated}} < p_T^{\mathrm{trigger}}$) for non-strange jets [14]. The studies described here investigate whether there are any indications that jets which produce strange particles are modified in heavy ion collisions. While jets producing a strange particle do not necessarily arise from a jet from a strange or anti-strange quark, the fraction of jets contribution to the di-hadron correlation arising from a strange or anti-strange quark is enhanced when a strange particle is produced as compared to di-hadron correlations without such a requirement. The goal of these studies is therefore to investigate whether there is any indication for a difference in the modifications seen by strange versus non-strange jets.

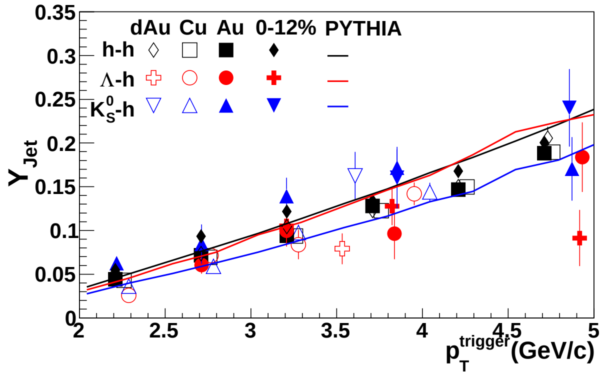

FIG. 2: The jet-like yield in $|\Delta\eta|$<0.78 as a function of p_T^{trigger} for h-h [14], K^0_S-h, and Λ-h correlations for 1.5 GeV/c $< p_T^{\mathrm{associated}} < p_T^{\mathrm{trigger}}$ in d+Au, Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}}$ = 200 GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [32] calculations using the Perugia 2011 tune [33].

CN: But this isn't necessarily so simple. We can also sample more charm jets by requiring strangeness - although I'd bet that this is suppressed substantially over strange jets at RHIC. And we also could sample more gluon vs quark jets. And then requiring a baryon could do different things entirely.

The jet-like yield as a function of p_T^{trigger} is shown in figure 2 for h-h [14], K^0_S-h, and Λ-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. Due to residual track merging effects, fits are used for Λ-h correlations in Cu+Cu collisions with $2.0 < p_T^{\mathrm{trigger}} <$ 3.0 GeV/c, in 0-12% Au+Au collisions for $3.0 < p_T^{\mathrm{trigger}} <$ 4.5 GeV/c, and in 40-80% Au+Au collisions for 2.0 $< p_T^{\mathrm{trigger}} <$ 4.5 GeV/c and for K^0_S-h correlations in 0-12% Au+Au collisions for $3.0 < p_T^{\mathrm{trigger}} < 3.5$ GeV/c and 40-80% Au+Au collisions for $2.0 < p_T^{\mathrm{trigger}} < 4.5$ GeV/c. There is no significant difference in the yields between the collision systems as observed earlier for h-h correlations in [14]. This includes no significant difference between results from Au+Au collisions in 40-80% and 0-12% central collisions, although there is a hint that the yield may be lower in central collisions, particularly at high p_T. The jet-like yield from K^0_S-h correlations is systematically above the jet-like yield from h-h correlations and the jet-like yield from Λ-h is systematically below the jet-like yield from h-h correlations, particularly in Au+Au collisions.

The difference between the different trigger particle species could be an indication that the requirement of a higher mass trigger particle reduces the amount of energy available for particle production at lower momenta. It could also indicate a difference in the shapes of the jet spectra leading to jets with a leading K^0_S, unidentified, hadron, or Λ. For instance, since higher energy jets lead to a higher associated yield, the difference between the trigger particles could indicate that jets which produce a

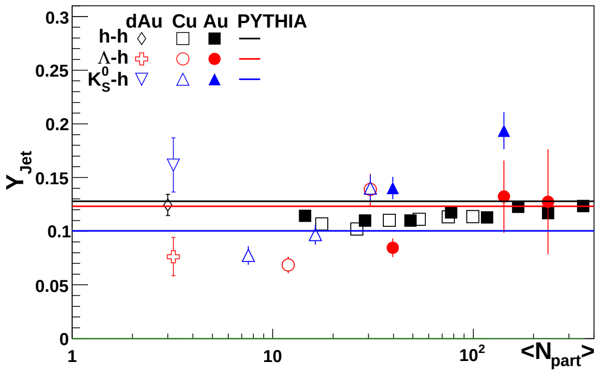

FIG. 3: Centrality dependence of the jet-like yield of h-h, K_S^0-h, and Λ-h correlations for $3 < p_T^{\rm trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{\rm associated} < p_T^{\rm trigger}$ in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [32] calculations using the Perugia 2011 tune [33]. Points have been shifted for visibility.

Λ are lower in energy on average than jets which produce a kaon.

Data are compared to PYTHIA calculations from the Perugia 2011 [33] tune. PYTHIA describes the trend of the jet-like yield well for all particle species. However, the trigger particle type ordering in PYTHIA is opposite that observed in the data, with the Λ-h jet-like yield comparable to the h-h jet-like yield, which is higher than K_S^0-h jet-like yield. If the dominant reason for the mass ordering in PYTHIA were that higher mass leading particles leave less energy for associated particle production, the mass ordering in PYTHIA would be the opposite. This indicates that the mass ordering in PYTHIA is dominanted by different jet energies contributing to the correlations and therefore indicates that PYTHIA either gets the shape of the strange jet spectra wrong or gets the underlying production mechanism for strange particles wrong. Several PYTHIA tunes have difficulty describing baryon production, in particular strange baryon production, and underestimates the K_S^0 spectra at high momenta [38–40]. These observations are consistent with underestimates of the strange quark spectra or with problems in the mechanism for strangeness production in PYTHIA.

Next the jet-like yields are studied as a function of collision centrality expressed in terms of number of participating nucleons ($N_{\rm part}$) calculated from the Glauber model [34]. The extracted jet-like yield as a function of $N_{\rm part}$ is shown in figure 3 for h-h [14], K_S^0-h, and Λ-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. While there is no centrality dependence in the jet-like yield of h-h correlations, there is a centrality dependence in the yields of the K_S^0-h and Λ-h correlations. This could be caused by many different factors. A difference between d+Au and A+A could be caused by nuclear effects. If strange jets experience more energy loss than light flavored jets, the jet-like yield associated with a strange trigger particle could increase with centrality even as the jet-like yield stays the same for light particles. If lower energy strange jets are suppressed more than lower energy light jets, the surviving spectrum of jets could be skewed towards higher energy jets and the corresponding jet-like yield would be higher. An increased yield could also come from a modification of fragmentation functions inside the medium, either from an enhancement in the production of lower energy particles in particles containing a strange jet or from an enhancement in the production of higher energy strange particles in light jets.

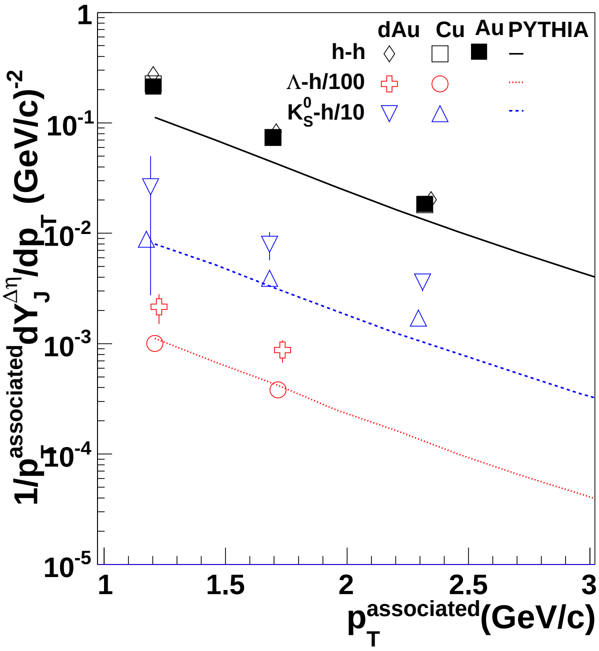

FIG. 4: The jet-like yield as a function of $p_T^{\rm associated}$ for h-h [14], K_S^0-h, and Λ-h correlations for $3 < p_T^{\rm trigger} < 6$ GeV/c in d+Au, Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}} = 200$ GeV. The data are compared to PYTHIA [32] calculations using the Perugia 2011 tune [33].

These data are compared to PYTHIA [32] calculations from the Perugia 2011 [33] tune. The PYTHIA tune used is not able to describe the trigger particle type ordering even for d+Au collisions, indicating that the production mechanism for strange particles in PYTHIA may need to be refined.

The jet-like yield as a function of $p_T^{\rm associated}$ is shown in figure 3 for h-h [14], K_S^0-h, and Λ-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. All yields are determined using bin counting. The Λ-h and K_S^0-h correlations are only shown for d+Au and Cu+Cu collisions since residual track merging made measurements in Au+Au collisions difficult. These data are compared to PYTHIA [32] calculations from the Perugia

2011 [33] tune. The fact that PYTHIA underestimates the jet-like yield in d+Au collisions could be because PYTHIA either does describe the spectrum of strange jets or their fragmentation well. A higher energy jet fragments into more particles, therefore, if the strange jet spectrum in PYTHIA is softer than in the data, PYTHIA will underestimate the yield associated with strange particles, even if the fragmentation of strange jets is correct. If PYTHIA assumes that a leading strange particle in a jet is formed at higher z=p_T^h/p_T^{jet} than in the data, PYTHIA will underestimate the yield associated with strange particles, even if the spectrum of strange jets is correct. Since neither the fragmentation nor the spectrum of strange jets are constrained well by data, it is also possible, or even probable, that both are wrong.

The Cu+Cu yields are lower than the d+Au yields. PYTHIA describes the Cu+Cu data better than the d+Au data; given the disagreement with the d+Au data, this is most likely a coincidence. The lower yield in the Cu+Cu spectrum could be due to either a modified strange jet spectrum in Cu+Cu data or modified fragmentation in the medium. At the LHC fragmentation functions are observed to be enhanced at lower momenta in A+A collisions relative to p+p [35, 36]. If this is also true at RHIC energies and at the lower jet energies sampled by with $3 < p_T^{\rm trigger} < 6$ GeV/c, the jet-like yield in Cu+Cu collisions would be expected to be enhanced rather than suppressed. Therefore the data in Figure 4 likely indicate modification of the jet spectrum in Cu+Cu collisions. Since no modification is observed for unidentified hadron triggers, this indicates greater modification of the strange jet spectrum than the non-strange jet spectrum.

For all data with identified strange trigger particles, PYTHIA is unable to describe the particle type ordering. The data indicate that there may be limitations in the models for strangeness production in jets, highlighting the need for both better models and more precise data.

B. Charged-V^0 correlations

The composition of the jet-like correlation can be studied using correlations with identified associated particles. The statistics of the d+Au data were limited and the Au+Au data set was limited by the presence of residual track merging. Therefore it was only possible to determine the composition of the jet-like correlation in Cu+Cu collisions. Since the jet-like correlation in h-h correlations has been shown to be consistent across collision systems [7, 14], the composition of the jet-like correlation is expected to be the same for all collision systems. These data are shown in figure 5 and compared to data from inclusive spectra in p+p collisions from STAR [37] and ALICE [38] and from PYTHIA [32] calculations from the Perugia 2010 and 2011 [33] tunes. The Cu+Cu data are consistent with the inclusive particle ratios from p+p, which is expected if the dominant production mechanism for strange particles at high momenta is mainly fragmentation. For each of the two PYTHIA tunes used, the inclusive ratio and the ratio in the jet-like correlation are comparable and lower than that observed in the data. The Perugia tunes, which improved the description of inclusive strangeness production at LHC energies, are further from the observed $\frac{\Lambda+\bar{\Lambda}}{2K_S^0}$ than Tune A. Several PYTHIA tunes have difficulty describing baryon production, in particular strange baryon production [38–40], indicating that this may be the reason for the discrepancy.The ratios are comparable between the jet-like correlation and the inclusive particle spectra in all PYTHIA tunes, as expected if both are dominated by fragmentation. This indicates that the jet-like correlation is dominantly produced by fragmentation in both the data and PYTHIA, however, the absolute values of the composition are limited by the difficulties with baryon production in PYTHIA

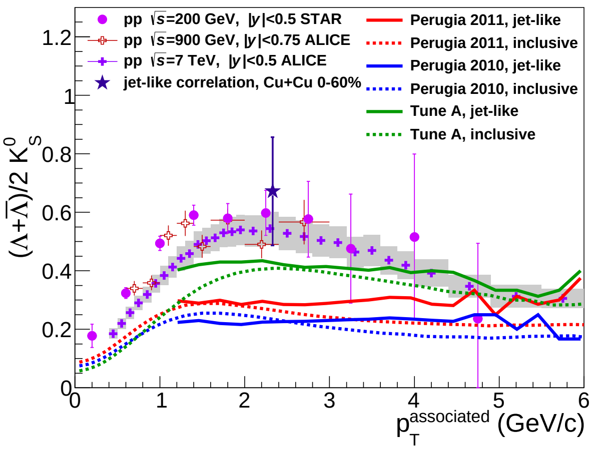

FIG. 5: Λ/K_S^0 ratio measured in the jet-like correlation in Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV for $3 < p_T^{\rm trigger} < 6$ GeV/c and $2.0 < p_T^{\rm associated} < 3.0$ GeV/c along with this ratio obtained from inclusive p_T spectra in p+p collisions.

V. CONCLUSIONS

We observe a trend for mass ordering of the yield of identified strange trigger particles in d+Au, Cu+Cu, and Au+Au collisions. This mass ordering is not the same ordering observed for the jet-like yield in PYTHIA. The $\frac{\Lambda+\bar{\Lambda}}{2K_S^0}$ ratio for the jet-like correlation is comparable to that observed in p+p collisions, providing further evidence that this correlation is dominantly produced by fragmentation. In PYTHIA, this ratio is comparable to the inclusive ratio. PYTHIA is unable to describe either the inclusive ratios or the ratio in the jet-like correlation. This indicates that there may still be some problems with the production mechanism for strange particles in PYTHIA.

These studies provide motivation for future studies of strangeness production in jets. Larger data sets and data from collisions at higher energies could provide more robust tests of the strangeness production mechanism in $p+p$ collisions. Studies in $p+p$ would be essential in order to search for modifications of strangeness production in jets in heavy ion collisions.

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